

ELEC4631 Tutorial 4

1. Consider again the LTI system from Tutorial 2 Q5, with input-output relation satisfying the linear second order ODE

$$y^{(2)}(t) + 2y^{(1)}(t) - 3y(t) = u^{(1)}(t) - u(t).$$

with initial conditions $y(0) = 0$ and $y^{(1)}(0) = 1$.

- (a) Suppose that $u(0) \neq 1$. Show that there does *not* exist an initial state x_0 of the control canonical form that will satisfy the given initial conditions for y .
- (b) Find the initial condition x_0 of the observer canonical representation so that the given initial conditions for y are satisfied.

(**Remark:** this exercise shows explicitly that the control canonical form cannot reproduce the solutions of certain ODEs with certain initial conditions for y , while the observer canonical form always can).

2. Recall the RLC circuit from lecture 5 with $R = 1 \Omega$, $L = 0.5 \text{ H}$, and $C = 1 \text{ F}$ and no driving input. The state-space representation of the circuit was

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ -2 & -2 \end{bmatrix} x(t),$$

with $x(t) = (v_C(t), i(t))$ and $x_0 = (v_0, i_0)$.

- (a) Write down the solution $x(t) = e^{At}x_0$ of the autonomous system.
- (b) Using the elementary trigonometric identities $\cos(\alpha+\beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$, $\cos(\alpha + \frac{\pi}{2}) = -\sin \alpha$, and $\sin(\alpha + \frac{\pi}{2}) = \cos \alpha$, show that the solution for $v_C(t)$ and $i(t)$ can be expressed as on slide 26 of Lecture 3 Part 1.

(**Remark:** for Lecture 5 the solution was actually determined more directly but of course yielding the same result).

3. Examine the internal stability and asymptotic stability properties of a system $S = (A, B, C, D)$ with the following A matrices:

(a)

$$A = \begin{bmatrix} -1 & -3 \\ 1 & -1 \end{bmatrix}.$$

(b)

$$A = \begin{bmatrix} 15 & -16 \\ 12 & -13 \end{bmatrix}.$$

(c)

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -4 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & 2 & 0 \end{bmatrix}.$$

(d)

$$A = \begin{bmatrix} 1 & -8 & 0 & 0 \\ 1 & -3 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

4. Consider the system

$$S = \left(\begin{bmatrix} 0 & 3 \\ 1 & -2 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix}, [0 \ 1], 0 \right)$$

- (a) Is the system internally stable?
- (b) Is the system internally asymptotically stable?
- (c) Is the system controllable?
- (d) Is the system observable?
- (e) Show that the system is BIBO stable when initially at rest ($x_0 = 0$) but the output diverges in magnitude (i.e., $\lim_{t \rightarrow \infty} |y(t)| = \infty$) for initial states x_0 which is not of the form $x_0 = (c, -c)$ for some real constant $c \neq 0$.

5. Consider the system

$$S = \left(\begin{bmatrix} 5 & -14 \\ 2 & -6 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \end{bmatrix}, [-2 \ 7], 0 \right).$$

- (a) Show that the system is not internally stable.
- (b) Compute the transfer function of the system.
- (c) Show that the system is BIBO stable. Then show that the output magnitude remains bounded at all times $t \geq 0$ for any initial condition x_0 despite the system being internal unstable.